

Tribhuvan University
Institute of Science and Technology
2074

Bachelor Level / Third Year /Six Semester/Science
Computer Science and Information Technology-(CSc.363)
(Image Processing)

Full Marks: 60
Pass Marks: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt any ten questions.

1. Explain how you can convert an analog image into a digital image. How many images of size 2400X1600 with 256 gray levels can be stored in a 1024 MB storage space? (3+3)
2. Explain the Haar transform with a suitable example. (6)
3. Explain the power law transformations. Write the algorithm for the implementation of median filter in spatial domain. (2+4)
4. What do you mean by low pass filtering in digital image processing? Explain it with suitable example. Show how can you convert low pass filter to high pass filter with suitable block diagram. (3+3)
5. What do you mean by Lossy Predictive Coding? Explain it with a suitable block diagram. (6)
6. Explain the detail procedure for implementing Butterworth High Pass filter in Frequency domain. (6)
7. What is zooming? Explain the process of zooming by interpolation method. (6)
8. Explain in detail the region split and merge techniques for image segmentation. List the problems associated with region split and merge technique. (4+2)
9. Explain in detail the template matching using Correlation. (4+2)
10. What is a gradient filter? Explain the Sobel gradient filter in detail along with its algorithm for implementation. (1+5)
11. Write short notes on: (3+3)
 - a) Pattern Recognition and its applications
 - b) Neural Network